FUNCTIONAL REQUIREMENTS DOCUMENT

Fraud Detection System — Loan Applications

Version 1.0 | March 2026 | Solo Project

Project Name	Fraud Detection System — Loan Applications
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1. Purpose

This document translates the business requirements defined in the Business Requirements Document (FDS-BRD-001) into detailed functional and technical system specifications. It defines exactly what the Fraud Detection System must do — how it ingests data, processes it, scores it, and responds — so that any developer, tester, or auditor can implement and verify the system without ambiguity.

Every functional requirement in this document maps to one or more business requirements in the BRD. Any requirement not traceable to the BRD is out of scope for v1.0.

2. System Overview

The Fraud Detection System is a standalone scoring service that sits between the Loan Origination System (LOS) and the credit decision engine. It exposes a REST API that accepts loan application data, runs it through a validation and preprocessing pipeline, scores it using a trained ML classification model, and returns a fraud risk score, decision label, and SHAP-based explanation to the calling system.

The system is composed of five logical components:

- Ingestion Layer — accepts both real-time API calls (JSON) and batch uploads (CSV); performs initial format and schema validation.
- Preprocessing Pipeline — applies field validation, PII stripping, categorical encoding, and feature engineering as defined in the Data Dictionary (FDS-DD-001).
- Scoring Engine — loads the trained model from the model registry and produces a fraud probability score (0.00–1.00) per application.
- Decision and Explanation Layer — converts the probability to a 0–100 risk score, applies threshold rules to assign a decision label, and generates SHAP feature importance outputs.
- Audit and Monitoring Layer — writes a complete log entry for every scoring event and feeds performance metrics (fraud flag rate, PSI drift) to the monitoring dashboard.

3. Data Inputs

The following 29 fields constitute the complete input schema for the scoring system. All fields are mandatory at inference time with the exception of IsFraud, which is the training target and must not be passed as an input during live scoring.

#	Field Name	Data Type	Valid Values / Range	Required	Preprocessing Note
01	**Age**	Integer	18 – 80	Yes	Numeric; validate range before scoring
02	**Occupation**	Categorical	Teacher / Engineer / Clerk / etc.	Yes	One-hot encode; 8 unique values
03	**MaritalStatus**	Categorical	Single / Married / Divorced	Yes	One-hot encode
04	**Dependents**	Integer	0 – 10	Yes	Numeric; validate range
05	**ResidentialStatus**	Categorical	Own / Rent	Yes	Binary encode: Own=0, Rent=1
06	**AddressDuration**	Integer	0 – 50	Yes	Numeric; validate range
07	**CreditScore**	Integer	300 – 850	Yes	Numeric; validate range; high fraud relevance
08	**IncomeLevel**	Float	> 0	Yes	Numeric; used in loan-to-income ratio feature
09	**LoanAmountRequested**	Float	> 0	Yes	Numeric; used in loan-to-income ratio feature
10	**LoanTerm**	Integer	1 – 360	Yes	Numeric; validate range
11	**PurposeoftheLoan**	Categorical	home / personal / auto / education / travel	Yes	One-hot encode
12	**Collateral**	Categorical	Yes / No	Yes	Binary encode: Yes=0, No=1
13	**InterestRate**	Float	0.0 – 30.0	Yes	Numeric; validate range
14	**PreviousLoans**	Integer	0 – 20	Yes	Numeric; validate range
15	**ExistingLiabilities**	Float	>= 0	Yes	Numeric

16	**ApplicationBehavior**	Categorical	Normal / Rapid	Yes	Binary encode: Normal=0, Rapid=1; high fraud relevance
17	**LocationofApplication**	Categorical	Local / Unusual	Yes	Binary encode: Local=0, Unusual=1; high fraud relevance
18	**ChangeinBehavior**	Categorical	Yes / No	Yes	Binary encode: No=0, Yes=1
19	**TimeofTransaction**	Time	HH:MM (00:00 – 23:59)	Yes	Extract hour as integer 0–23 before scoring
20	**AccountActivity**	Categorical	Normal / Suspicious	Yes	Binary encode: Normal=0, Suspicious=1
21	**PaymentBehavior**	Categorical	On-time / Late / Default	Yes	Ordinal or one-hot encode; highest fraud share on Default
22	**Blacklists**	Categorical	Yes / No	Yes	Binary encode: No=0, Yes=1; triggers auto-reject pre-score
23	**EmploymentVerification**	Categorical	Verified / Unverified	Yes	Binary encode: Verified=0, Unverified=1
24	**PastFinancialMalpractices**	Categorical	Yes / No	Yes	Binary encode: No=0, Yes=1; highest correlation with target
25	**DeviceInformation**	Categorical	Mobile / Laptop / Tablet	Yes	One-hot encode
26	**SocialMediaFootprint**	Categorical	Yes / No	Yes	Binary encode: Yes=0, No=1
27	**ConsistencyinData**	Categorical	Consistent / Inconsistent	Yes	Binary encode: Consistent=0, Inconsistent=1
28	**Referral**	Categorical	Referral / Online / etc.	Yes	One-hot encode
29	**IsFraud**	Binary	0 / 1	Target	Target variable only; not passed as input at inference time

4. Functional Requirements

Priority legend: Must Have = mandatory for go-live | Should Have = important but not blocking | Nice to Have = desirable if time permits.

ID	Requirement	Priority	Maps to BRD	Notes
FR-01	System shall accept application data either as a batch CSV file or as a real-time JSON payload via REST API.	Must Have	BR-01	Both ingestion modes required
FR-02	System shall validate all 29 input fields against defined type, range, and allowable-value rules before any scoring is attempted.	Must Have	BR-01	Invalid records routed to error queue
FR-03	System shall apply the pre-processing pipeline — encoding, feature engineering, PII exclusion — to every validated record before passing it to the model.	Must Have	BR-01, BR-08	Encoding per Data Dictionary
FR-04	System shall generate a fraud probability score in the range 0.00–1.00 for each application using the trained ML model.	Must Have	BR-01	Score stored alongside application ID
FR-05	System shall convert the raw probability to a 0–100 risk score and apply the threshold rules defined in Section 6 to determine the decision label.	Must Have	BR-01	Low / Medium / High / Auto-Reject
FR-06	System shall auto-reject any application where the Blacklists field is Yes, prior to model scoring, without consuming model resources.	Must Have	BR-04	Pre-scoring rule; bypasses FR-03/FR-04
FR-07	System shall flag all High-risk applications (score >= 75) and route them to the manual review queue with a review-required status.	Must Have	BR-05	Queue entry created automatically
FR-08	System shall generate a SHAP-based explanation listing the top 5 contributing features and their direction of influence for every application that receives a Medium or High risk label.	Must Have	BR-02	Output stored and surfaced to reviewer
FR-09	System shall write a complete audit log entry for every scoring event, containing: application ID, timestamp, all input feature values, computed score, decision label, and SHAP output.	Must Have	BR-03	Log must be tamper-proof and append-only
FR-10	System shall return the score, decision label, and SHAP output to the calling system (LOS API) within 3 seconds for 95% of requests under normal load.	Must Have	BR-06	p95 latency SLA
FR-11	System shall strip PII fields (Name, PAN, Contact, Email, Address) from any input payload before the record enters the feature engineering pipeline.	Must Have	BR-08	PII never reaches the model
FR-12	System shall handle a missing or null value in any non-critical field by applying the defined imputation strategy and logging a data quality warning.	Must Have	BR-01	Missing critical fields route to manual review
FR-13	System shall route any application where the model service is unavailable to the manual review queue and trigger an IT alert; it must never auto-approve on a system failure.	Must Have	BR-06	Fail-safe: default to manual review

FR-14	System shall expose a /health endpoint that returns the model version, last retrain date, and current service status.	Must Have	BR-07	Used by monitoring and LOS integration
FR-15	System shall generate a weekly performance summary report covering: total applications scored, fraud flag rate, false-positive rate, and model drift indicators (PSI).	Should Have	BR-09	Automated; delivered to configured recipients
FR-16	System shall support loading and activating a new model version without requiring a full system restart, using a versioned model registry.	Should Have	BR-10	Enables monthly retraining pipeline
FR-17	System shall support concurrent scoring of at least 50 simultaneous API requests without degradation in response time.	Should Have	BR-06	Concurrency requirement
FR-18	System shall provide an admin endpoint to update the fraud score threshold (default 75) without requiring a code deployment.	Should Have	BR-05	Configurable threshold
FR-19	System shall retain all audit logs for a minimum of 7 years in compliance with applicable financial-record retention regulations.	Must Have	BR-03	Retention policy requirement
FR-20	System shall provide a confidence interval alongside each fraud score to assist reviewer prioritisation within the High-risk band.	Nice to Have	BR-11	Score calibration output

5. System Behaviour — Scoring Logic

The following steps describe the complete processing sequence for every application received by the system.

Step 1 — Pre-Score Blacklist Check

Before any preprocessing or model scoring occurs, the system checks whether the Blacklists field is Yes. If yes, the application is immediately auto-rejected, an audit log entry is written, compliance is notified, and no further processing takes place. This step is defined in FR-06.

Step 2 — Input Validation

All 29 input fields are validated against their defined data type, range, and allowable-value rules (FR-02). Records with invalid values are routed to the error queue with a descriptive validation error. Records with missing non-critical fields proceed to imputation (FR-12). Records missing critical fields are routed to the manual review queue.

Step 3 — Preprocessing Pipeline

Validated records pass through the preprocessing pipeline (FR-03): PII fields are stripped, categorical fields are encoded per the Data Dictionary, derived features are calculated (Section 8), and the complete feature vector is assembled.

Step 4 — Model Scoring

The assembled feature vector is passed to the trained model. The model returns a fraud probability between 0.00 and 1.00 (FR-04). This probability is converted to a 0–100 risk score.

Step 5 — Threshold Decision

The risk score is evaluated against the threshold rules defined in Section 6 (FR-05, FR-07). A decision label is assigned: Auto-Reject, High, Medium, or Low. High and Medium applications trigger SHAP explanation generation (FR-08).

Step 6 — Response and Logging

The score, decision label, and SHAP output are returned to the LOS via the API response (FR-10). Simultaneously, a complete audit log entry is written containing the application ID, timestamp, all input features, score, decision label, and SHAP output (FR-09). If the audit log write fails, scoring is halted for that record and IT is alerted.

6. Alert & Threshold Rules

The default fraud score threshold is 75 out of 100. This value is configurable by an admin without code redeployment (FR-18). The table below defines the full decision matrix.

Score Range	Risk Level	System Action	Description
0 – 40	**Low**	Auto Proceed	Application passes through the origination pipeline without any additional review step.
41 – 74	**Medium**	Soft Flag — Monitor	Application is flagged as medium risk. SHAP explanation is generated. Loan officer is notified and may approve with documented justification.
75 – 100	**High**	Hard Flag — Review	Application is immediately routed to the manual review queue. Cannot be approved without a reviewer sign-off.
N/A	**Auto-Reject**	Blocked Pre-Score	Applicant appears on a blacklist. Application is rejected before model scoring. Compliance is notified automatically.

7. Non-Functional Requirements

ID	Requirement	Target	Notes
NFR-01	API response time	p95 < 3 seconds	Measured per scoring request under normal load; logged by monitoring
NFR-02	System uptime	>= 99.5%	Measured monthly; excludes scheduled maintenance windows
NFR-03	Audit log retention	Minimum 7 years	Logs stored in append-only storage; deletion requires compliance sign-off
NFR-04	Concurrent request handling	>= 50 simultaneous	System must not degrade below p95 SLA under this load
NFR-05	Model scoring throughput	>= 200 req / minute	Sustained throughput requirement for batch processing windows
NFR-06	Data encryption at rest	AES-256 minimum	All audit logs and model artefacts encrypted at rest
NFR-07	Data encryption in transit	TLS 1.2 or above	All API communication over HTTPS

NFR-08	PII exposure	Zero tolerance	No PII field may appear in model inputs, logs, or SHAP outputs
NFR-09	Model retraining frequency	Monthly minimum	Automated pipeline; triggered earlier if PSI drift threshold is breached
NFR-10	New model activation downtime	Zero downtime	Model versioning must support hot-swap without service interruption

8. Feature Engineering Specification

In addition to the 28 encoded input features, the preprocessing pipeline creates the following derived features before scoring. These features must be included in the final model feature vector.

Derived Feature	Calculation	Type	Rationale
LoanToIncomeRatio	LoanAmountRequested / IncomeLevel	Float	Compound risk signal; outperforms either feature alone in EDA
HourOfApplication	Hour extracted from TimeofTransaction	Integer	0–23; captures late-night application risk pattern
TimeOfDayBand	Binned hour: Morning / Afternoon / Evening / LateNight	Categorical	One-hot; handles non-linear time patterns
DebtToIncomeRatio	ExistingLiabilities / IncomeLevel	Float	Measures over-leverage; high DTI is a fraud signal
HighRiskOccupation	1 if Occupation in [Unemployed, Self-employed] else 0	Binary	Derived from EDA occupation fraud rate analysis
IsRapidAndUnusual	1 if ApplicationBehavior=Rapid AND LocationofApplication=Unusual	Binary	Interaction feature; compound high-risk signal
BlacklistOrMalpractice	1 if Blacklists=Yes OR PastFinancialMalpractices=Yes	Binary	Combined strong fraud indicator

9. API Specification

The system exposes a REST API (JSON over HTTPS) for integration with the Loan Origination System and any other authorised consumer. All endpoints require authentication via API key passed in the request header.

Endpoint	Method	Input	Response	Description
POST /score	POST	JSON payload	200 OK + score response	Submit a single application for real-time fraud scoring
POST /score/batch	POST	CSV or JSON array	202 Accepted + job ID	Submit a batch of applications for asynchronous scoring
GET /score/{id}	GET	Application ID	200 OK + full result	Retrieve the stored score and SHAP output for a past application

GET /health	GET	None	200 OK + status JSON	Returns model version, last retrain date, and service status
GET /audit/{id}	GET	Application ID	200 OK + audit record	Retrieve the full audit log entry for a given application ID
PUT /threshold	PUT	JSON config	200 OK	Update fraud score threshold values without code redeployment (admin only)

Sample Scoring Request (POST /score)

{ "application_id": "APP-00123", "Age": 35, "CreditScore": 640, "IncomeLevel": 45000, "LoanAmountRequested": 120000, "ApplicationBehavior": "Rapid", "LocationofApplication": "Unusual", "Blacklists": "No", "PastFinancialMalpractices": "Yes", ... }

Sample Scoring Response

{ "application_id": "APP-00123", "fraud_probability": 0.83, "risk_score": 83, "decision": "High", "action": "Manual Review Required", "top_features": [{"feature": "PastFinancialMalpractices", "impact": "+0.31"}, {"feature": "ApplicationBehavior", "impact": "+0.22"}, ...], "timestamp": "2026-03-20T14:35:02Z" }

10. Error & Edge Case Handling

Scenario	System Response	Outcome	Ref
Missing CreditScore	Flag for manual review; log data quality warning; do not score	Manual review queue	FR-12
Missing mandatory field (any)	Route to manual review; log warning; alert data team	Manual review queue	FR-12
Applicant on Blacklist	Auto-reject before scoring; notify compliance; write audit log	Rejected	FR-06
Invalid field value (out of range)	Reject record; return validation error response; log details	Error queue	FR-02
Model service unavailable	Route to manual review; trigger IT alert; never auto-approve	Manual review queue	FR-13
API timeout (> 10 seconds)	Default to manual review; log timeout event; alert IT	Manual review queue	FR-13
Duplicate application ID	Reject with duplicate error; do not score; log warning	Error queue	FR-02
PII field detected in payload	Strip PII; log security warning; proceed with cleaned record	Score with cleaned data	FR-11
Malformed JSON / CSV payload	Return 400 Bad Request; log malformed payload ID; alert sender	Error queue	FR-01

Inconsistent data (ConsistencyinData = Inconsistent)	Score normally; SHAP will surface as top feature; flag to reviewer	High flag likely	FR-08
Score exactly on threshold (= 75)	Treated as High risk; routed to manual review queue	Manual review queue	FR-07
Audit log write failure	Halt scoring for that record; retry 3 times; alert IT on failure	Scoring paused	FR-09

11. Assumptions & Constraints

Assumptions

- The Loan Origination System can make outbound REST API calls and consume JSON responses; no changes to the LOS UI or core logic are required.
- The model artefact and feature pipeline are versioned together in the model registry; a mismatch between model version and pipeline version is treated as a deployment error.
- All 29 input fields are available at the point of application submission; partial records are the exception, not the norm.
- The admin threshold update endpoint is access-controlled and available only to authorised risk management personnel.

Constraints

- The system must never auto-approve an application as a result of a system failure, timeout, or missing data. The fail-safe default is always manual review.
- PII must not appear in any log, audit entry, SHAP output, or API response under any circumstance.
- v1.0 supports a single model at a time in production. A/B testing across two live models simultaneously is deferred to v1.1 (FR-20 equivalent).
- The fraud threshold default of 75 may be adjusted by an admin but must remain within the range 50–90; values outside this range are rejected by the configuration endpoint.